Blood group classification using Hybrid Transformer XAI and FedAvg technique

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Abstract

Blood group identification plays an important role in medical diagnostics as it helps in safe blood transfusion. Traditionally this work is done manually in laboratories through serological tests though victims blood samples. These conventional method is accurate but time consuming and impractical in urgent conditions or resources are limited. In recent years, machine learning and deep learning techniques are explored to automate the classification tasks. Also most models are black box models which makes even more difficult to understand how model predictions are generated. This proposed work provides an optimal approach for classifying blood groups with the help of fingerprint images using hybrid transformer model specifically SWIN Transformer with EfficientNet (CNN), here CNN EfficientNet model effectively captures the texture and fine details, whereas transformer model SWIN which uses attention mechanism focus on global contextual relationships. In addition to these X-ai techniques also implemented using Grad-CAM to visualize important regions that influence model's decisions. Furthermore Federated learning implemented to enable decentralized training. The proposed work presents a complete framework for performance, interpretability and privacy learning for blood group classification.

Keywords: SWIN, Transformer, EfficientNet, CNN, Grad-CAM, Federated

1 Introduction

Accurate identification of blood group plays an important role, since the consequences of this are significant, whether it's blood transfusion, organ transplant or emergency medical situations. Classifying the blood group with the help of fingerprints makes it accurate and reliable as fingerprints are fundamental biometric features, uniquely identifying individuals through patterns formed by ridges and valleys on their fingertips. These patterns are developed in the early birth stage of individual and remains unchanged throughout the life making them a reliable means. The three primary fingerprint patterns includes loops, whorls and arches. Beyond these basic patterns fingerprints also contains minutiae points such as bifurcations, ridge endings, islands which adds additional uniqueness to individual fingerprint.

There are several studies that shows the correlation between these fingerprint patterns and certain blood group types. Our method utilizes this natural characteristic for classifying the blood group. The dataset we used consist of eight ABO types (A+, A-, B+, B-, O+, O-, AB+, AB-) blood group fingerprints. By utilizing the effectiveness of deep learning models specifically CNN and Transformer, we can effectively capture these features and make the prediction reliable. The CNN identifies intricate fingerprint patterns by learning features from ridge endings, bifurcations, and minutiae points in the image while Transformer can focus on distinct regions in an image also referred as global features using it's attention mechanism which can improve feature representation. Even though both approaches are effective alone, using only one not capturing the information needed for accurate classification.

Keeping performance is important, many medical AI systems lacks interpretability as most of the models are black boxes, meaning the operator can not easily understand how decision are made. In medical sector lack of transparency limits adoption in real world. Explainable Artificial Intelligence (XAI) methods helps in solving this problem. In this work, Gradient weighted Class Activation Mapping (Grad-CAM) is used to generate heatmaps that highlights image regions that are influencing model decision. Including XAI increases the reliability and interpretability of the system.

Another issue in medical AI is data privacy. This data is sensitive and are not shared publicly which makes it difficult to collect large datasets. Sharing raw data may not always be possible due to ethical and legal constraints. Federated learning provides solution to this problem by allowing models to train over multiple local datasets without transferring raw data. This process allows collaborative learning while preserving privacy.

2 Literature Review

Khandelwal et al. [1]. The author introduced a non-invasive way to identify blood groups through fingerprint images, concentrating on the textures and patterns. Their method relied on image processing and machine learning for classification. This study improves upon fingerprint-based blood group prediction by implementing CNNs for feature extraction. SaiKrishna et al. [2]. The author suggested to use an automated blood group identification system based on image processing for non-invasive antigen

feature extraction and classification. This study involves extracting antigen characteristics from images of blood samples to forecast blood types. Ansari et al. [3]. The author introduced a system called Blood Snap for detecting blood groups using image processing techniques. In their study, blood samples were captured as images and analyzed using image-based feature extraction methods to classify different blood groups. However, the model did not make use of deep learning techniques or advanced optimization methods, which limited the improvement of accuracy. Still, the study provided helpful insights. C. Sivamurugan et al. [4] focuses on a new way of identifying blood groups using fingerprint images with the help of deep learning techniques. In this study, Convolutional Neural Networks (CNNs) were used to learn and analyze unique fingerprint patterns. Since fingerprints are easy to collect and do not require blood samples, this method aims to provide a fast and non-invasive option compared to traditional blood testing methods. Nihar et al. [5] introduced a different approach for identifying blood groups using fingerprint analysis. Their study focused on understanding the relation between fingerprint ridge patterns and different blood types. In their work, pre-trained deep learning models such as AlexNet and LeNet-5 were used to classify fingerprint images.The study highlights the potential of combining biometrics and deep learning for blood group identification and provides a foundation for further research in this area. Swathi P et al. [6]. In this study, they used Convolutional Neural Networks (CNNs) to analyze detailed fingerprint image features. The model was trained using different fingerprint patterns with the aim of predicting the blood group of an individual without using blood samples. The results of the study showed that the proposed deep learning model achieved a moderate accuracy. Fatima et al. [7]. The author proposed an automatic system for identifying blood groups along with a notification alert feature. In their work, image processing techniques were used to analyze blood sample images and determine the blood group. Once the blood group was detected, the system could send alerts or notifications to inform the users. By combining blood group detection with an alert system, the proposed approach helps in faster communication and better efficiency. S. A. Shaban et al. [8] proposed a method to identify blood groups using image processing techniques. In their study, around 600 images of blood samples collected from different laboratories were used for analysis. The authors applied techniques such as CLAHE for image enhancement, Otsu's thresholding for segmentation, and various MATLAB-based tools to process the images. Blood group classification was mainly done by detecting clumps in the blood samples, separating RGB color components, and counting objects present in the images. Vijaykumar et al. [9] presented a new method to predict ABO blood groups using fingerprint images with the help of a Convolutional Neural Network (CNN). In their work, the authors designed and optimized a CNN model to study fingerprint patterns and relate them to blood groups. This approach avoids the need for blood samples, making the process simple, quick, and non-invasive compared to traditional methods. P. N. Vijaykumar et al. [10] carried out a study to predict blood groups using fingerprint features with the help of machine learning methods. In their work, techniques such as Multiple Linear Regression using the OLS method and Gabor filters were used to analyze fingerprint images. The study was conducted on a dataset of 82 students, where important fingerprint features like minutiae points, ridge count,

and the ratio between ridge thickness and valley thickness were extracted and studied. Hatfaludi et al. [11] studied how large pre-trained models can be used together with federated learning systems. In their research, they created a clear classification that shows different ways these pre-trained models can be adjusted and used in a federated setup. They explained how such models can be fine-tuned or adapted when the training data is spread across many devices instead of being stored in one place.They also talked about several problems that can happen in this type of system. In addition, not all devices have the same computing power, which creates resource limitations. The study concluded that using foundational models with federated learning can help the model perform better on new data, while still protecting user privacy. Guidotti et al. [12] conducted survey on explainability methods for black-box models. In their study, they compared two main types of approaches: model-agnostic methods, which can be used with any model, and model-specific methods, which are designed for a particular model type. They explained how these techniques are applied in different areas, such as visual explanations for image-based models, text-based explanations for NLP systems, and counterfactual explanations that show what changes in input could lead to a different prediction. The authors also discussed how to measure the quality of explanations. In the end, the paper pointed out the existing gaps in evaluating explainable AI methods and suggested possible directions for improving research in this field. Cooray et al. [13]conducted review on deep federated learning, focusing specifically on deep neural networks which are trained in federated environment. In their work, they studied how to use this approach in different fields such as health centers, computer vision, NLP, etc. and what challenges are faced during optimization for stable and efficient training process. At the end author concluded that federated learning given good performance in many applications and also suggested that future work focus on faster model convergence and lower communication overhead. Ribeiro et al. [14]introduces LIME, which stands for Local Interpretable Model-agnostic Explanations for explaining machine learning models. This works by creating a simple and easily interpretable model around a single prediction of complex model. By this we are trying to understand why model gave a specific output. They tested LIME on variety of tasks such as text classification and image recognition and showed that even complex black box models can interpreted this way, without loosing any performance of the model. Al et al. [15] conducted review of research on federated learning and its use in areas like healthcare, IoT, finance, and smart systems. In their study, they explained how federated learning makes it possible for different users or organizations to train a shared model without sending their raw data to a central server. They also pointed out some important challenges in federated learning. The authors highlighted that better and more adaptive aggregation methods are needed so that the global model can perform well even when client datasets are very different from each other. Liu et al. [16] studied federated learning by looking at it from the angle of multi-party computation (MPC) and secure joint training. In their paper, they explained different privacy protection methods used in federated systems. These methods include secure aggregation, homomorphic encryption, and secret sharing. The authors also grouped federated learning systems into three main types: centralized, decentralized, and hierarchical architectures. Their study showed that when MPC techniques are combined

with federated learning, the chances of data leakage during model updates can be reduced to a large extent. Riedel et al. [17] conducted a comparative study of open-source federated learning frameworks such as TFF (TensorFlow Federated), PySyft and Flower. In their study, comparitive study is based on scalability, flexibility, convenience and communication at deployment time. The author discussed advantages and disadvantages of each framework. Alazab et al. [18] conducted review of federated learning, specifically on aggregating methods, system design and implementation tools. In thier work, author explained different algorithms and their modified versions. they also compared other federated learning development frameworks to understand practical issues during deployment and concluded that major challenges are communication and Non-IID data when dealing with federated learning. Ye et al. [19] studied HFL which stands for Hetrogeneous Federated Learning, where different clients has different data distribution, model sturcture and computing capacity. In their work, because of the differences in data, device and computation, training global model becomes complex. The author grouped HFL into three different categories which includes data level solution, model level solution and system level solution and concluded that proper handling of statistical and system hetrogenity is crucial to make federated systems reliable, scalable, and efficient for practical scenarios. Arya et al. [20] worked on a toolkit and classification system of explainable AI methods. In this, they grouped XAI techniques based on different factors. The author tested many explanation methods on synthetic and real world problems and concluded that no single XAI technique performs best in every situation, thus usage of an explanation method depends upon the task and needs. They further suggested the importance of standard benchmarks to properly measure how effective and meaningful interpretability methods. Adadi et al. [21] conducted survey on XAI methods. In their work, they worked on two types of approaches which are post-hoc methods and intrinsic methods. They reviewed XAI techniques that are used in different tasks like classification, clustering and deep learning applications. The author also discussed different ways to evaluate explanation and gaps in measuring how useful these explanations are. They also addresses how XAI is applied in different areas and concluded that interpretability is more important in real world applications. Lundberg et al. [22] introduced a method called SHAP, which stands for SHapley Additive exPlanations. This framework is based on ideas from game theory and is used to explain complex machine learning models. In their work, they used Shapley values from cooperative game theory to calculate how much each feature contributes to a specific prediction. The authors tested SHAP on different types of models, such as gradient boosting trees and deep neural networks.

3 Methodology

3.1 Data Collection

In this study, we used Finger Print Blood group dataset publicly available on Kaggle. It consists of fingerprint images categorized into eight blood group classes: A+, A-, B+, B-, AB+, AB-, O+ and O-. Each class contains multiple fingerprint samples collected under controlled conditions. Below given Fig. 1 shows the distribution of blood group samples in the dataset.

All images are stored in class-wise directories, where each folder represents a specific blood group. The dataset was organized into two folders for training and testing while developing model and evaluation results. In this, the number of samples is not exactly the same for all blood group classes. Some classes have more samples, while others have fewer. This difference can make the model focus more on the classes with higher samples and ignore the smaller ones. To solve this problem, we used the class weight technique during training.
In this method, higher importance is given to the classes with fewer samples, and lower importance is given to the classes with more samples. This helps the model pay equal attention to all blood group categories. As a result, the model learns in a more balanced way and improves its performance for minority classes as well.

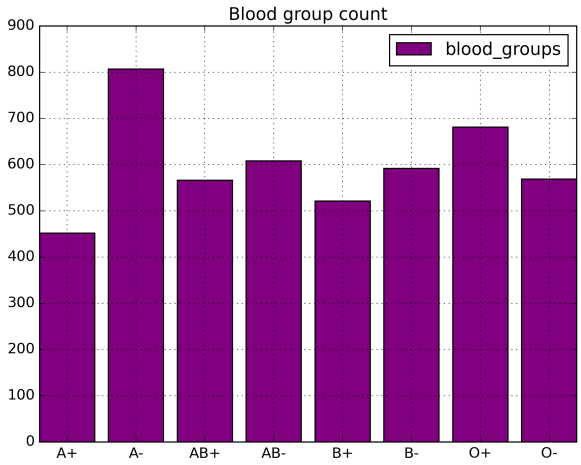

Fig. 1: Blood group distribution chart.

3.2 Data preprocessing

For each fingerprint image, the following preprocessing steps are applied sequentially, and numeric features are extracted:

1. Image enhancement to improve ridge visibility and reduce noise.
2. Gabor filtering to enhance ridge orientation patterns.
3. Otsu thresholding to convert the image into binary form.
4. Skeletonization to reduce ridge structures to single-pixel width.
5. Removal of short branches to eliminate spurious ridge fragments.
6. Minutiae extraction to detect ridge endings and bifurcations.
7. Border removal and pruning to discard unreliable minutiae near image boundaries.

8. GLCM texture feature computation from the enhanced image.

From these operations, a total of 15 numeric features are extracted. These include statistical minutiae features and texture-based GLCM features.

3.3 Hybrid Combination

Fig. 2 shows the result of an experiment where we compared two CNN modes these are ResNET and EfficientNet with four transformer models which includes SWIN, ViT, DeiT, CaiT, for finding out the best hybrid model.
All these models are finetuned for 30 epochs with early stopping and final results is SWIN alone able to achieve the 90% of validation accuracy and early stopped at epoch 19 outperforming all the other models behind it EfficientNet model performed the best with validation accuracy of 88% early stopped at epoch 24, which helps in identifying the final hybrid fusion model combo using EfficientNet with SWIN transformer, rest all the models scored less than these models. But ResNet50 gives a tough competition to EfficientNetB3 model but not able to achieve as good accuracy as EfficientNetB3 making it the best choice.

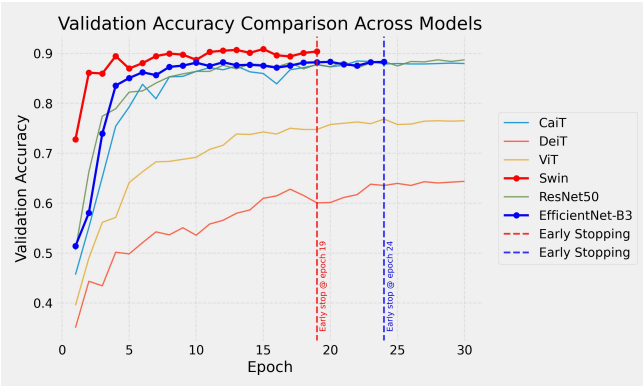

Fig. 2: Comparison of CNN and Transformer models.

3.4 Proposed Work

The proposed approach utilizes a hybrid model for blood group classification using fingerprint images. The model combines deep learning representations with biometric features. The main idea behind this approach is that different techniques capture different types of features, and combining them can improve the overall classification performance.
Let X be a fingerprint image and $y \in \{1, 2, \ldots, 8\}$ be its corresponding blood group label. The objective is to learn a mapping function

$$f(X) \to y$$

where the output belongs to one of the eight blood group classes. Instead of relying solely on raw image pixels, we define a multimodal feature representation as:

$$F = [F_{\text{cnn}}, F_{\text{swin}}, F_{\text{num}}]$$

where:

- F_{cnn} represents deep convolutional features,
- F_{swin} represents transformer-based global features,
- F_{num} represents handcrafted numerical fingerprint features.

The job of the classifier is to learn a mapping, which increases the representation capacity of the model.
Formally, this can be expressed as:

$$g(F) \to y$$

where F is the multimodal feature representation and y is the predicted blood group label.

3.4.1 CNN branch

CNN branch is based on the EfficientNet model and is responsible for learning the spatial hierarchy of features. The early layers detect edges and ridges, the middle layers detect local patterns, and the deep layers capture complex fingerprint patterns. Mathematically, convolution is defined as:

$$Y(i,j) = \sum_m \sum_n X(i-m, j-n)\, K(m,n)$$

where K is the convolution kernel.
After extracting the final feature map, global average pooling is applied. This produces a 96-dimensional feature vector, computed as:

$$F_{\text{cnn}} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} M(i,j)$$

where H and W are the height and width of the feature map M.

3.4.2 SWIN branch

The proposed model uses Swin Transformer, which leverages the attention mechanism, while the first branch of the hybrid model uses CNN, which works on convolution

operations.
Mathematically, self-attention is computed as:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

The attention mechanism allows the Swin Transformer to capture relationships between distant regions of a fingerprint image.It effectively captures global ridge orientation and structural consistency.
The transformer output is then projected to a 256-dimensional vector, representing high-level structural information:

$$F_{\text{swin}} \in \mathbb{R}^{256}$$

3.4.3 Numeric branch

Along with the deep learning features extracted from CNN and SWIN Transformer branches, fingerprint analysis methods are used to extract additional handcrafted features from each training image. These features include:

1. Log count of ridge endings and bifurcations
2. Std. deviation and mean of x-coordinates of ridge endings and bifurcations
3. Std. deviation and mean of y-coordinates of ridge endings and bifurcations
4. Log ratio of endings to bifurcations
5. Four GLCM texture features

Ridge endings and bifurcations are fundamental biometric characteristics. Minutiae are detected using ridge skeleton analysis. GLCM features measure statistical texture properties, such as contrast and homogeneity. The purpose of these features is to provide a statistical description of the fingerprint.

3.4.4 Latent Encoding

The extracted handcrafted features are passed through an encoder network, and the auto-encoder learns a lower-dimensional latent representation of these features. This helps remove noise, reduce redundancy, and produce a compact biometric representation.
Mathematically,

$$z = \text{Encoder}(x)$$

where

$$x \in \mathbb{R}^{15}, \quad z \in \mathbb{R}^{12}.$$

After the projection layer, the numeric branch produces a 128-dimensional feature vector. This ensures that the numeric features are on a compatible scale with the deep

learning features extracted previously:

$$F_{\text{num}} \in \mathbb{R}^{128}.$$

3.4.5 Feature Fusion and Classification layer

The fusion of 96 CNN features, 256 transformer features, and 128 encoded numeric features produces a 480-dimensional representation. This allows the classifier to learn interactions between local texture, global structure, and biometric statistics. Mathematically,

$$R_{\text{fusion}} = R_{\text{cnn}} \oplus R_{\text{swin}} \oplus R_{\text{num}}$$

where,

$$R_{\text{fusion}} \in \mathbb{R}^{480}.$$

The final concatenated features are input into a fully connected classifier and class weights are applied to handle imbalance classification along with cross-entropy loss function.
The classifier operations can be written as:

$$h = \text{GELU}\left(W_1 F_{\text{fusion}} + b_1\right)$$

$$\hat{y} = \text{Softmax}\left(W_2 h + b_2\right)$$

The cross-entropy loss:

$$L = -\sum_{i=1}^{C} y_i \log \hat{y}_i$$

where:

- F_{fusion} is the 480-dimensional concatenated feature vector,
- W_1, W_2 are the weight matrices,
- b_1, b_2 are the bias,
- h is the hidden representation after GELU activation,
- C is the number of classes,
- y_i and $\hat{y}_i$ are the predicted probabilities for class i.

3.5 Training Pipeline

The fingerprint dataset is organized in a directory-based structure, where each blood group is represented by a separate folder. Images inside each folder correspond to that class. The dataset is divided into two parts: training (80%) and validation (20%) sets. For each training sample, the following components are calculated:

- *img_eff:* Image tensor for the CNN branch
- *img_swin:* Image tensor for the Transformer branch
- *numeric_feat:* Numeric feature vector
- *label:* Class index

Two different image preprocessing pipeline are applied because CNN and Transformer takes inputs of different dimensions. Below given Table 1. describes the transformation for EfficientNet and SWIN.

Table 1: Transformations for CNN and SWIN branch

Transformation	**EfficientNet**	**SWIN**
Resize	300×300	224×224
Horizontal flip	0.5	0.5
Rotation	±10	±10

The feature vector created while preprocessing data is scaled using a pretrained scaler and converted into tensor and this tensor is later passed to the hybrid model as the numeric input branch.

The hybrid model contains three parallel branches: CNN, SWIN, and Numeric branch. In this hybrid model, we used the *EfficientNetB3* model pretrained on ImageNet. This CNN branch processes the 300×300 resized image, and the final convolution feature map is passed through convolution, batch normalization, and average pooling layers. This produces the *img_eff* vector of 96 dimensions.

The second branch is the *SWIN* transformer `swin_tiny_patch4_window7_224`). This branch uses a 224×224 resized image. Unlike CNN, it uses the attention mechanism to capture long-range relationships within the fingerprint image. The output embedding is passed through a linear layer, layer normalization, and GELU activation to produce the *img_swin* vector of 256 dimensions. This captures the global structure and ridge flow information.

The last branch is the *Numeric* branch, which takes 15 handcrafted features as input. An auto-encoder reduces these features to 12 dimensions to remove redundancy and noise. The projection layer then expands the representation to a 128-dimensional *numeric_feat* vector. The outputs from all three branches are concatenated and fused into a single 480-dimensional vector:

$$F_{\text{fusion}} = [\textit{img_eff} \oplus \textit{img_swin} \oplus \textit{numeric_feat}] \in \mathbb{R}^{480}$$

The classifier consists of a linear layer, GELU activation, and Dropout layer. The final output layer predicts one of the eight blood group classes. The training is performed in two stages:

1. Frozen backbone training: In this stage, the backbones of EfficientNet and SWIN are frozen, and only the classifier and numeric layers are trained for 12 epochs. This stabilizes the fusion layer and trains the classifier on pretrained features.

2. Partial fine-tuning: In this stage, the last stage of the SWIN transformer is trained with a smaller learning rate for 30 epochs. Early stopping is implemented with patience value of 5, monitoring the best validation accuracy.

3.6 Results

After training, the hybrid model achieves an accuracy of 92.94%, with a precision of 92.97% and recall of 93.16% on validation dataset.
For evaluating the model's discriminative capability, ROC curve analysis was performed using a one-vs-rest strategy for all eight blood group classes. The ROC curve is generated by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at different decision thresholds.

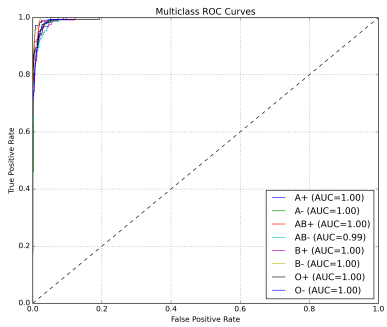

Fig. 3: ROC curve of Hybrid model

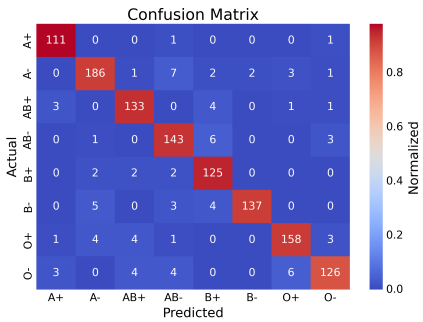

Fig. 4: Confusion matrix

The Area Under the Curve (AUC) measures class separability. An AUC value of 1.0 indicates perfect separability between the target class and the remaining classes, meaning the model can effectively differentiate positive samples from negative samples. A slightly lower AUC of 0.99 is observed for the AB− blood group, which still represents reliable discrimination capability. Fig. 3 shows that the model achieves high sensitivity while maintaining an extremely low false positive rate across all classes.
To further evaluate the classification performance of Hybrid model confusion matrix is constructed. It provides a detailed class-wise distribution of correct and incorrect predictions. The model an overall accuracy of 92.94%, indicating that majority of fingerprint samples are correctly classified. Fig. 4 represents the models confusion matrix, where most of samples are correctly predicted which shows the reliability of the model.

4 XAI Integration

To uncover the black box of deep learning models and understand how decision are made by the hybrid model, Gradient Weight Class Activation Mapping (Grad-CAM)

is implemented. This helps in visualizing which regions in the fingerprint image contributes more in final prediction. In our work, Grad-CAM is implemented on CNN branch, specifically on the final convolution feature map before global average pooling and IG implemented on SWIN by accumulating gradients.

4.1 Grad-CAM on CNN

For a given fingerprint image first forward pass is performed through hybrid model, the predicted blood group score is selected and gradients of particular class are computed with respect to the final CNN feature maps. These gradients are globally averaged for calculating weights and later multiplied with corresponding feature maps. A weighted sum is also calculated and ReLU activation is applied to keep only positive regions. Finally, heatmap is superimposed on the original fingerprint image, which produces a visual explanation showing the regions that contributes the most in prediction.

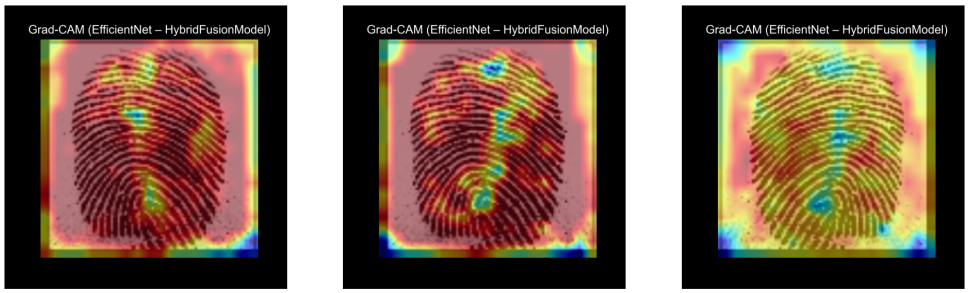

Fig. 5: Grad-CAM on different layers of CNN.

Fig. 5 represents heatmap shows highlighted regions in warm colors (red/yellow) and less important regions in cool color (blue). Warm color represents the high contribution to classification and blue region indicate low contribution. In our experiment, Grad-CAM visualization showed that model primarily focus on core ridge structures, minutiae-dense region and central fingerprint patterns. This shows that CNN learns discriminative ridge patterns instead of noise. Although final decision based on CNN, SWIN and numeric features, Grad-CAM analysis shows CNN captures local ridge details.

4.2 IG on SWIN

Integrated gradients (IG) implemented using *captum* library. Unlike Grad-CAM which is limited to convolution layers, IG can explain transformer based models like SWIN. It measures how much each input feature contributes in prediction by accumulating gradients along a path from baseline input to actual input. As we know, SWIN uses self-attention instead of convolution operation thus Grad-CAM is not reliable for full transformer explanation.
In our experiment, IG computed for input image of size (224x224) passed to SWIN

along with numeric feature vector, thus it provides attribution for both visual and numeric modality. Instead of using single gradient, it integrates gradients over multiple small steps *n_step=50* in our implementation.

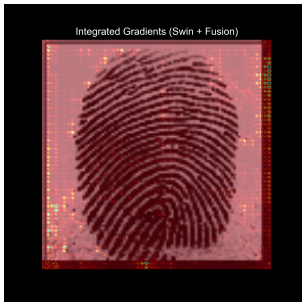

Fig. 6: IG on SWIN.

Fig. 6 represents the results of IG on SWIN. Although the CNN branch provides clear spatial localization using Grad-CAM, interpretability of the SWIN branch is comparatively less localized. This is due to the architecture difference between convolutional and attention mechanism models.
CNNs maintain explicit spatial feature maps, the SWIN uses window-based self-attention and hierarchical patch merging, which distribute importance across multiple regions. Therefore, IG attribution for the SWIN appears smoother and more globally distributed. It is important to clarify that comparative lower localization clarity in transformer explanation, Transformer are inherently more difficult to interpret because attention weights are not always equal to feature importance, multiple attention heads contributes in prediction process, hierarchical merging alters spatial mapping. Therefore, interpretability techniques for transformers are still in research area.

5 FedAvg Hybrid model

Fingerprint-based blood group classification involves sensitive biometric information. Centralized training requires collecting all fingerprint images into a single server, which increases the risk of privacy leakage. To address this issue, a *Federated Learning (FL)* framework was implemented. It allows multiple clients to collectively train a global model without sharing raw data. Instead, only model parameters are exchanged. This will ensures data privacy along with benefits of distributed data.
The federated system consist of a central server, multiple clients and hybrid model as base architecture. Each client contains raw fingerprint images, numeric feature extraction pipeline, copy of the hybrid model, local optimizer, and training routine. The server initializes the global model and distributes weights to clients. Each client trains locally and returns updated weights to the server, where aggregation is performed to update the global model. The dataset is divided into multiple subsets, where each subset represents a clients. Each client trains only on its local fingerprint images. This depicts the real world scenario where data is stored across different institutions. The

federated learning process follows the standard Federated Averaging (FedAvg) algorithm. The server initializes the Hybrid model with pretrained weights . The global model weights are sent to all participating clients. Each client performs the following steps:

1. Load local dataset
2. Extract numeric features
3. Apply preprocessing and scaling
4. Train of hybrid model for a specific number of epochs
5. Update model weights using backpropagation

After local training, the updated model weights are sent to the server. The server aggregates client weights using weighted averaging:

$$W_{\text{global}} = \sum_{k=1}^{K} \frac{n_k}{n} W_k$$

where, W_k is the weights from client k, n_k is the number of samples available to client k, n is the total samples, and K is the total clients.
This will ensures that those clients which has more data contribute proportionally more to the global update. The aggregated weights replace the previous global model weights. This process is repeated for multiple rounds until convergence.

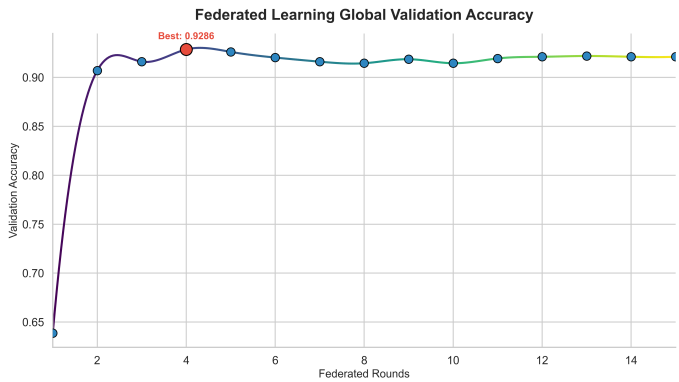

Fig. 7: Global validation accuracy in Federated rounds.

In our experiment, the federated framework trains the complete hybrid model, including all three branches. Since the CNN and SWIN backbones are partially frozen, most updates occur in the projection layers, numeric branch, and classifier layer. Training achieves 99% of centralized accuracy while ensuring data privacy, with convergence in under five rounds. Fig. 7 shows how the global model accuracy varies after each federated round. The best accuracy achieved is 92.84% in just four epochs, after which the model fluctuates around 92% accuracy. Although the performance

remains competitive while preserving data privacy, convergence is more sensitive compared to centralized training, and model divergence is possible if client updates vary significantly.

6 Conclusion and Furture scope

In this work, a hybrid deep learning framework was proposed for fingerprint-based blood group classification. The system integrates CNN and transformer architecture, and numeric features into a single model. This hybrid strategy allows the model to capture local structural patterns along with global contextual information, which leads to an overall validation accuracy of $\approx 93\%$. and $\approx 95\%$. of training accuracy for classification. The results confirms that fingerprint pattern contains features that are related with blood group. Integrating XAI along with Federated learning technique strengthens the model by providing visual evidence of features and ensure that sensitive fingerprint data remains decentralized which makes overall system reliable for practical use cases.

Although the proposed hybrid model achieves good validation accuracy but still, a larger and mnore diverse dataset is still suggested for imporoving the generalization ability of model. This would help evaluate robustness in real world scenarios. Better transformer architecture can be explored to imporove global feature learning. For XAI point of view SWIN transformer capture contextual information, its interpretability is still limited when compared to CNN. In addition, adaptive federated optimization methods can be explored for faster convergence on Non-IID data distribution. Futhermore, combining fingerprint features with other biometrics trains may improve classification performance and model reliablity.

References

[1] Thakre, D., Panpatte, M., Khandelwal, R.: Non invasive technique for fingerprint-based blood group identification. In: Proceedings of the Fifth International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT), pp. 1–7 (2025)

[2] Reddy, G.C., SaiKrishna, G., Micheal, A.A.: Automated blood group detection system using image processing for non-invasive antigen feature extraction and classification. In: Proceedings of the 3rd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT), pp. 2199–2208 (2025)

[3] Ansari, T., Jamadar, T., Sonde, A., Jamkhandikar, I.R.: Blood snap–blood group detection using image processing. In: Proceedings of the IEEE 4th International Conference on ICT in Business Industry & Government (ICTBIG), pp. 1–4 (2024)

[4] Sivamurugan, C., Perumal, B., Siddarthha, Y., Murthi, V.K., Sankar, Y.N., Varun, Y.: Enhanced blood group prediction with fingerprint images using deep

learning. In: Proceedings of the 3rd International Conference on Automation, Computing and Renewable Systems (ICACRS), pp. 1091–1097 (2024)

[5] Nihar, T., Yeswanth, K., Prabhakar, K.: Blood group determination using fingerprint. MATEC Web of Conferences **392**, 01069 (2024)

[6] P, S., K, S., Horadi: Fingerprint based blood group using deep learning. International Journal of Advanced Research in Science, Communication and Technology, 699–708 (2024)

[7] Dewal, B., Rathi, M., Memon, Y., Fatima, K.: A framework for automatic blood group identification and notification alert system. Sir Syed University Research Journal of Engineering & Technology **13**(2), 74–84 (2023)

[8] Shaban, S.A., Elsheweikh, D.L.: Blood group classification system based on image processing techniques. Intelligent Automation & Soft Computing **31**(2), 817–834 (2022)

[9] Patil, V.N., Ingle, D.R.: A novel approach for abo blood group prediction using fingerprint through optimized convolutional neural network. International Journal of Intelligent Systems and Applications in Engineering **10**(1), 60–68 (2022)

[10] Vijaykumar, P.N., Ingle, D.R.: A novel approach to predict blood group using fingerprint map reading. In: Proceedings of the 6th International Conference for Convergence in Technology (I2CT), pp. 1–7 (2021)

[11] Hatfaludi, C.-A., Serban, A.: Foundational models and federated learning: Survey, taxonomy, challenges and practical insights. PeerJ Computer Science **11** (2025)

[12] Guidotti, R., *et al.*: Survey of methods for explaining black box models. ACM Computing Surveys **55**(9), 1–40 (2025)

[13] Cooray, L., et al.: Deep federated learning: A systematic review of methods, applications, and challenges. Frontiers in Computer Science (2025)

[14] Ribeiro, M.T., Singh, S., Guestrin, C.: Why should i trust you? explaining the predictions of any classifier. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 1135–1144 (2024)

[15] Mohammed, T.S., Al-Libawy, H.: Literature review of federated learning in various applications, challenges, and emerging research directions. Journal of University of Babylon for Engineering Sciences (2024)

[16] Liu, F., et al.: A survey on federated learning: A perspective from multi-party computation. Frontiers of Computer Science **18** (2024)

[17] Riedel, P., et al.: Comparative analysis of open-source federated learning frameworks — a literature-based survey. International Journal of Machine Learning and Cybernetics (2024)

[18] Alazab, A., et al.: A systematic review of federated learning: Challenges, aggregation methods, and development tools. Journal of Network and Computer Applications **225** (2023)

[19] Ye, M., et al.: Heterogeneous Federated Learning: State-of-the-art and Research Challenges. arXiv preprint (2023)

[20] Arya, N., et al.: One explanation does not fit all: A toolkit and taxonomy of ai explainability techniques. Transactions on Machine Learning Research **2023** (2023)

[21] Adadi, A., Berrada, M.: Peeking inside the black-box: A survey on explainable artificial intelligence (xai). IEEE Access **8**, 182384–182415 (2023)

[22] Lundberg, S., Lee, S.: A unified approach to interpreting model predictions. In: Advances in Neural Information Processing Systems, vol. 35, pp. 4765–4774 (2022)